University of Southern Mississippi

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CSC 430: Senior Design Proj

Aircraft Maintenance Anomaly Detection System

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Senior Project Proposal

Project Overview

Modern aircraft are capable of producing enormous amounts of logs and information. Scheduled maintenance approaches are good at preventing failures, but many mechanical issues are preceded by small anomalies in the logs, which are hard to recognize for human operators, especially when dealing with thousands of records from aircraft fleets.

This project aims to design and implement an *Aircraft Maintenance Anomaly Detection System* to recognize early warning signs from historical trends in aircraft maintenance records.

This system is not intended to be used for autonomous decision-making for aircraft maintenance. Instead, it is intended to be used as an aid for making decisions, which is in accordance with real-world practices in aviation. This system is intended to be used as an aid for making decisions in accordance with real-world practices in aviation.

Problem Statement

Problem: Aircraft maintenance data are typically complex, high-volume, and temporally distributed across various systems. Current maintenance analysis methods are unable to recognize small, initial-stage deviations that lead to maintenance activities or equipment degradation, especially when such deviations do not relate to known failure mechanisms.

Goal: The goal of this project is to design a system that can automatically analyze the maintenance logs for the aircraft, as well as other related data, to recognize abnormal patterns that may signify high risk or mechanical problems, yet remain interpretable and relevant for human maintenance workers.

Why We Chose This Project

We chose this project because aircraft safety is not only dependent on advanced mechanical and aerospace engineering but also on the extent to which humans can understand complex aircraft data and respond appropriately. In most cases involving aircraft failure due to maintenance-related causes, it is not a single factor but a combination of complex trends, anomalies, and degradations that are hard to identify.

The project presents a great chance to utilize some of the basic concepts of computer science, including data analysis and system design. By emphasizing explainability and responsible system design, this project is aimed at assisting humans rather than replacing them, which is a realistic and ethical practice in the aviation industry.

Conclusion

The Aircraft Maintenance Anomaly Detection System will be an example of a realistic and impactful problem in aviation using computer science. This problem is relevant to aviation safety and is relevant in terms of both industry practices and academic rigor, while at the same time being feasible in terms of an undergraduate senior project.